

Iterative Policy Evaluation for Cats Example

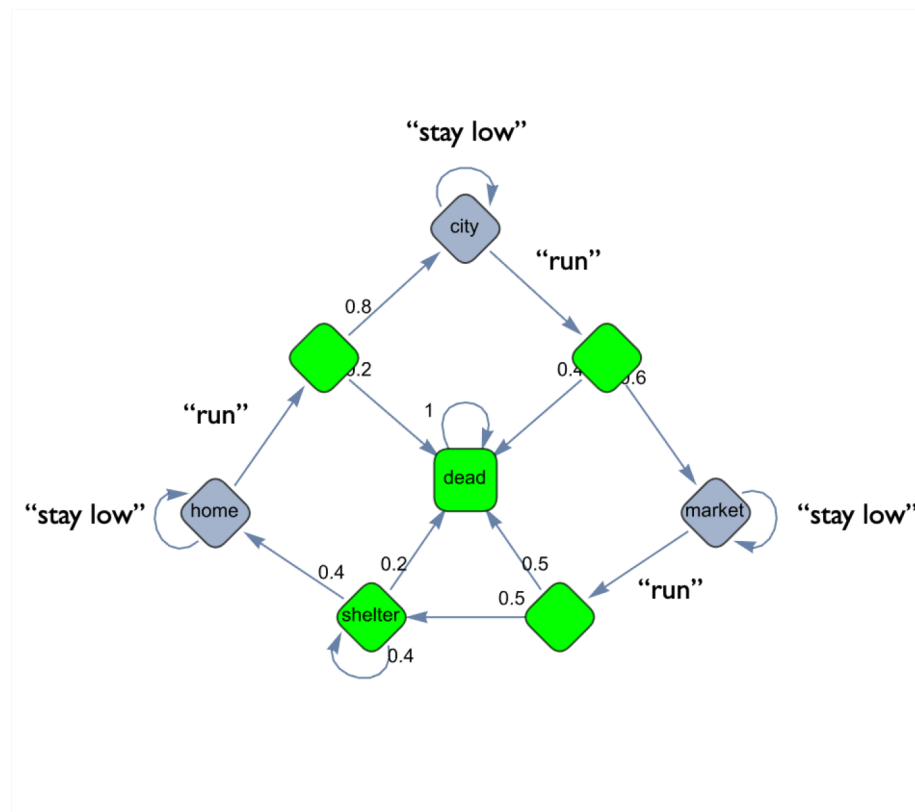
Essentially, we are iterating the Belman Equation, reading it as an update equation

$$v_{\pi}^{t+1} \leftarrow R^{\pi} + \gamma P^{\pi} v_{\pi}^t$$

which arises from

$$v_{\pi}(s) \leftarrow E[R_{t+1} + \gamma v_{\pi}(S_{t+1}) \mid S_t = s]$$

The relevant transition properties are best read off from this diagram



We first have to fix a policy. For a simple example, we make both actions equally likely, ie

$$\pi(r \mid m) = 0.5; \pi(s \mid m) = 0.5$$

We then also need an initial value array. We pick

```
In[*]:= V = Range[5]
Out[*]:= {1, 2, 3, 4, 5}
```

to be able to see effects clearly.

Let's calculate the update of the market value .

First we need the expected return for each action.

```
In[*]:= Rrun = 0.5 rewards[[5]] + 0.5 rewards[[2]]
Out[*]:= 0.1
```

```
In[*]:= Rstay = rewards[[4]]
Out[*]=
0.7
```

The weighted average for this policy then is

```
In[*]:= R = 0.5 * Rrun + 0.5 * Rstay
Out[*]=
0.4
```

Likewise, we need the expectation of the value of the followup state for each possible action.

```
In[*]:= Vrun = 0.5 V[[5]] + 0.5 V[[2]]
Out[*]=
3.5
```

Note that the above are not the probabilities for the policy but the transition probabilities for a given action

```
In[*]:= Vstay = V[[4]]
Out[*]=
4
```

Finally, we need to set a discount factor

```
In[*]:= γ = 0.9;
Out[*]=
```

The weighted average for this policy then is

```
In[*]:= V[[4]] = 0.5 * (Rrun + γ Vrun) + 0.5 * (Rstay + γ Vstay)
Out[*]=
3.775
```